Continuous Assessment Card

Department of Physiology... Dental college/ Unit...

Students name... Roll no. ....

Session... Year............................... Batch................................

Date of starting.. Date of ending ....

Card 1: (General Physiology & Blood)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature &
1.	Definition, goal & importance of physiology. Homeostasis: definition, major functional systems, control systems of the body	10		
2.	The cell: Organization, cell membrane, cell organelles and their functions.	10		
3.	The cell membrane transport: active & passive transport, exocytosis & endocytosis. Membrane potential: definition and basic physics of membrane potential. Resting membrane potential Nerve Action potential & propagation of action potential.	10		
4.	Neuromuscular junction: skeletal muscle contraction and relaxation. Transmission of impulse from nerve ending to the muscle fiber.	10		
5.	Composition & functions of blood, Plasma proteins: Origin, normal values & functions.	10		
6.	RBC: normal count, morphology, functions, erythropoiesis. Hemoglobin: types, functions & fate. Red blood cell indices: PCV, MCV, MCH & MCHC. Anaemia: definition & classification Polycythemia: definition & type. Jaundice: definition & classification	10		
7.	WBC: classification with normal count, morphology & functions. Leucocytosis, leucopenia .	10		
8.	Platelets: normal count & functions. Hemostasis: definition & events Coagulation: definition, blood clotting factors . Mechanism of coagulation Anticoagulant: name. Bleeding disorder: thrombocytopenic purpura & hemophilia. Tests for bleeding disorder: bleeding time, coagulation time and prothrombin time.	10		
9.	Blood grouping: ABO & Rh system, hazards of blood transfusion & Rh incompatibility.	10		

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Card No.—2: (Cardiovascular System)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Properties of cardiac muscle. Junctional tissues of the heart. Generation of cardiac impulse & its conduction in the heart.	10		
2.	Cardiac cycle: definition, events, pressure & volume changes during different phases of cardiac cycle. Heart sounds : type, characteristics ECG : definition and interpretations	10		
3.	Functional classification of blood vessels Local & humoral control of blood flow in the tissues. Exchange of fluid through the capillary membrane.	10		
4.	SV, EDV, ESV: definition & factors affecting them. Cardiac output : definition and factors affecting cardiac output. Venous return: definition & factors affecting. Heart rate: factors affecting & regulation. Tachycardia, bradycardia Pulse: definition, characteristics	10		
5.	Peripheral resistance: definition & factors affecting. Blood pressure: definition, types, measurement & regulation of arterial blood pressure.	10		

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Card 3: (Respiratory System)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Respiration: definition, mechanism. Pulmonary & Alveolar ventilation. Pulmonary volumes and capacities Dead space: physiological & anatomical Lung function tests : name & significance	10		
2.	Composition of atmospheric, alveolar, inspired and expired air. Respiratory unit and respiratory membrane. Diffusion of Gases through the respiratory membrane.	10		
3.	Transport of Oxygen & Carbon dioxide in blood.	10		
4.	Respiratory centers: name, location & functions. Nervous & chemical regulation of respiration. Hypoxia: definition, types Cyanosis: definition & types. Definition of dyspnea, hypercapnea & periodic breathing.	10		

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Card 4: (Endocrinology & Reproductive system

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Endocrine glands: name Hormones: definition, classification, mechanism of action	10		
2.	Hypothalamic hormones (releasing, inhibitory), Pituitary hormones (anterior & posterior): name and functions.	10		
3.	Thyroid hormones: biosynthesis, transport and functions.	10		
4.	Parathyroid hormone: functions, mechanism of action. Calcium & Phosphate metabolism, tetany.	10		
5.	Adrenocortical hormones: name, functions and regulation of secretion.	10		
6.	Hormones of endocrine pancreas: functions, consequences of hyperglycaemia and hypoglycaemia	10		
7.	Introduction to male reproductive physiology, spermatogenesis, functions of testesterone	10		
8.	Introduction to female reproductive physiology Menstrual cycle, ovarian and endometrial cycle Definition of ovulation, menstruation, menarche & menopause. Functions of oestrogen and progesterone	10		

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Card 5: (Nervous system & special sense)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Functional organization and functions of major levels of central nervous system (CNS). Neuron: definition, parts, types Nerve fiber: classification, effects of injury to the nerve fiber Synapse: properties & synaptic transmission Neurotransmitters: definition, types & functions	10		
2.	Sensory systems of the body: Sensory receptors, reflex, reflex arc, general/somatic senses, Ascending tracts/sensory pathways: name & function	10		
3.	Descending tracts/ motor pathways: name, function. Upper motor neuron and lower motor neuron: definition, effect of lesion.	10		
4.	Function of cerebellum, thalamus, hypothalamus and basal ganglia	10		
5.	Normal body temperature, site of measurement, sources of heat gain, channels of heat loss, regulation of body temperature in hot and cold environment.	10		
6.	Autonomic Nervous system: functional organization, functions.	10		
7.	Basic concept of vision: visual receptor, visual pathway, refractive errors	10		
8.	Basic concept of hearing: auditory apparatus, receptor, Deafness.	10		
9.	Smell: receptor and pathway. Taste: receptors, modalities of taste sensation and pathway.	10		

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Date of starting... Date of ending ....

Card 6: Physiology Practical

(I hear and I forget, I see and I remember, I do and I understand)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Use of microscope, laboratory equipment, collection (venous & capillary) of blood.	10		
2.	Preparation & staining of blood film for differential count of WBC with interpretation and analysis of result.	10		
3.	Estimation of haemoglobin with interpretation and analysis of result.	10		
4.	Determination of packed cell volume (PCV), Calculation of MCV, MCH & MCHC with interpretation and analysis of result.	10		
5.	Estimation of ESR by Westergren method with interpretation and analysis of result.	10		
6.	Determination of bleeding time, clotting time with interpretation and analysis of result.	10		
7.	Determination ABO & Rh blood groups with interpretation and analysis of result.	10		
8.	Clinical examination of radial pulse / respiratory rate.	10		
9.	Measurement of normal blood pressure.	10		
10.	Recording of ECG	10		
11.	Recording of oral & axillary temperature.	10		
12.	Observation of reflexes: superficial & deep.	10		

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Session.. Year............................... Batch...............................

Date of starting.. Date of ending ....

Card No- 1. Biophysics and Biomolecules

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Introduction of biochemistry, acid, base, pH, pK, buffer.	10		
2.	Define solutions, standard solution, crystalloid and colloid. Define & classify isotope and state its biomedical importance	10		
3.	Define & classify carbohydrates, protein & lipids	10		
4.	Define & classify enzymes, coenzymes, cofactors, isoenzymes. Describe the factors affecting enzyme activity.	10		
5.	Define steroids and sterols. Describe the sources & biomedical importance of cholesterol. Define & classify lipoproteins and mention their biomedical importance.	10		

Card No- 2. Food, nutrition and vitamins

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Basic concepts of Nutrient, food, diet, balanced diet, essential dietary components, calculation of calorie requirement. BMR, BMI, SDA and RDA.	10		
2.	Dietary fibers, nutritional importance of carbohydrate, lipid & protein.	10		
3.	Minerals (macro & micro), trace elements, common nutritional disorders, PEM, BMI. obesity, iron metabolism and its deficiency, iodine deficiency.	10		
4.	Water soluble vitamins: definition, classification, sources, functions, RDA and deficiency disorders.	10		
5.	Fat soluble vitamins: definition, classification, sources, functions, RDA and deficiency disorders and toxicity.	10		

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Date of starting.. Date of ending ....

Card 3 : GIT, Bioenergetics and Metabolism

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Phygiologic anatomy of gastrointestinal tract (GIT). Enteric nervous system. Movements of the GIT. Local hormones of GIT: name, functions & regulation of secretion	10		
2.	Digestive juices: composition, functions and regulation of secretion	10		
3.	Digestion and absorption of carbohydrate, protein and fat	10		
4.	Functions of liver and liver function tests.	10		
5.	Bioenergetics: biological oxidation, high energy phosphates, oxidative phosphorylation. Metabolism: definition, phases; anabolism, catabolism	10		
6.	Carbohydrate metabolism: glycolysis, fate of pyruvate, TCA cycle, definition of gluconeogenesis, glycogenesis & glycogenolysis.	10		
7.	Lipid metabolism: lipolysis, fate of Acetyl-CoA, ketone bodies, ketosis , lipoproteins & their importance, cholesterol metabolism	10		
8.	Protein metabolism: Amino acid pool, Transamination, Deamination. Source & fate of ammonia, ammonia intoxication.	10		

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Session... Year.............................. Batch..................................

Date of starting.. Date of ending ....

Card 4 : (Kidney, Body fluid, electrolytes and acid base balance)

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Body fluids compartments, fluid intake and output chart.	10		
2.	Kidney: functions of kidneys. Nephron : structure, types, parts & functions	10		
3.	Mechanism of urine formation Glomerular filtration rate (GFR), determinants of GFR	10		
4.	Reabsorption and secretion by the renal tubules Renal threshold, tubular load & plasma load.	10		
5.	Mechanism of formation of concentrated & dilute urine. Diuresis: definition & types	10		
6.	Acidification of urine Micturition reflex	10		
7.	Acid-Base Balance- origin of acids & bases, maintenance of static blood pH.	10		
8.	Serum Electrolytes- Serum electrolytes & their reference ranges. Functions, regulations, hypo & hyper states of serum Na+, K+, Ca++ & PO4--	10		

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Card No-5 . Clinical biochemistry and clinical endocrinology

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Clinical biochemistry: S I unit, laboratory hazards, Sample collection, photometry.	10		
2.	Clinical enzymology: lipid profiles of blood and their clinical importance.	10		
3.	Diagnosis of diabetes mellitus. OGTT and its interpretation, definition of IGT, IFG and HbA1c.	10		
4.	Thyroid function tests and their interpretation.	10		
5.	Common liver function tests (LFT). Jaundice.	10		
6.	Renal function tests and their interpretation.	10		

Card No-6. Fundamental of molecular biology and genetics

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Definition of Nucleic acids, nucleotides, DNA. transcription and translation	10		
2.	Definition of Gene, Genetic code, mutation, Genotype, Phenotype, RNA.	10		
3.	Recombinant DNA technology, PCR, concept of biotechnology	10		

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Card 7: Biochemistry Practical

Sl. No.	Name of item	Full Marks	Marks Obtained	Remarks (signature & Date)
1.	Laboratory etiquette, common laboratory mishaps and its prevention.	10		
2.	Identification of laboratory glass wares & equipments, photoelectric colorimeter	10		
3.	Preparation of solution from supplied solute, solvent and standard solution.	10		
4.	Estimation of blood glucose	10		
5.	Benedict's test, heat coagulation test and Rotherus test in urine.	10		
6.	Interpretation of results of blood glucose, serum urea, serum creatinine, serum total protein and serum bilirubin	10		

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